

FAIRHOPE 3 NE, AL, USA

WMO: 722370

Lat: 30.5485N

Lon: 87.8757W

Elev: 95

StdP: 14.65

Time Zone: -6.00 (NAC)

Period: 06-19

WBAN: 63869

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	26.2	29.8	13.6	11.0	31.5	18.8	14.2	36.9	14.7	64.6	12.9	63.6	1.9	N/A	0.319

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.8	91.9	77.8	90.3	77.6	88.9	77.4	80.5	88.0	79.9	87.3	79.1	86.3	4.3	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
78.5	148.4	84.8	77.8	144.9	84.3	77.1	141.6	83.7	44.2	88.3	43.3	87.3	42.6	86.4	83.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.7	10.9	9.5	19.9	95.1	3.4	2.4	17.4	96.8	15.4	98.2	13.5	99.6	11.0	101.4
			18.4	82.5	4.0	0.8	15.5	83.0	13.1	83.5	10.9	84.0	7.9	84.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	66.4	49.7	54.0	59.2	65.1	72.4	79.0	80.4	80.3	76.8	67.5	57.2	54.2
	DBStd	12.93	10.23	9.88	8.23	6.20	4.74	3.04	2.56	2.75	3.95	7.58	8.26	9.85
	HDD50	322	134	68	20	1	0	0	0	0	0	1	26	72
	HDD65	1774	483	323	213	72	9	0	0	0	0	64	257	353
	CDD50	6307	124	181	304	455	696	871	942	941	805	543	242	203
	CDD65	2286	8	17	32	76	239	421	477	476	356	141	23	20
	CDH74	20178	7	28	111	489	2059	4155	4706	4494	2876	1109	105	39
	CDH80	7100	0	0	2	29	536	1591	1875	1818	1005	238	6	0
(14) Wind	WSAvg	3.7	4.7	4.8	4.8	4.2	3.3	2.7	2.5	2.4	2.9	3.6	3.8	4.6
(15) Precipitation	PrecAvg	67.0	5.5	5.5	5.8	4.6	4.9	6.5	8.2	6.9	6.0	3.5	4.7	4.6
	PrecMax	94.1	16.3	14.4	11.6	20.5	13.7	18.5	28.6	17.2	24.1	13.1	12.0	9.0
	PrecMin	41.0	0.9	0.8	0.4	0.1	0.3	0.7	0.8	1.3	0.6	0.0	0.6	1.7
	PrecStd	12.8	3.4	3.0	2.8	4.3	3.6	4.1	5.2	3.6	4.7	3.0	3.0	1.7
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	75.0	77.3	79.3	83.0	90.1	93.6	93.3	93.8	92.1	87.7	80.4	77.3
		MCWB	67.8	68.8	66.6	71.6	74.0	77.6	78.6	77.8	76.7	75.7	70.8	71.6
	2%	DB	72.3	74.4	77.1	80.6	86.9	90.8	91.3	91.7	89.3	85.0	76.9	75.0
		MCWB	67.8	67.8	68.1	70.5	72.7	77.0	78.5	78.4	75.9	73.8	68.9	69.7
	5%	DB	69.8	71.5	74.9	78.8	85.0	88.9	89.5	90.1	87.5	82.8	74.7	72.7
		MCWB	66.0	66.7	67.2	69.3	73.0	77.4	78.3	78.2	75.9	72.3	67.7	68.4
	10%	DB	67.1	69.2	72.3	77.1	83.2	87.3	88.1	88.4	85.9	80.3	72.1	70.1
		MCWB	63.8	65.0	66.0	67.9	72.4	77.0	77.7	77.9	75.6	70.9	65.3	66.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	70.4	71.4	71.5	75.3	77.0	81.1	81.5	82.1	80.1	78.5	73.9	73.5
		MCDB	72.9	75.0	75.5	79.9	84.0	88.8	89.0	89.6	88.0	83.1	77.3	76.2
	2%	WB	68.7	69.9	70.4	73.4	75.9	80.0	80.3	80.8	78.7	76.8	72.0	71.6
		MCDB	71.3	73.3	74.2	77.7	83.0	87.5	88.0	88.1	86.0	81.8	75.3	74.1
	5%	WB	66.8	68.1	69.0	72.2	75.0	79.0	79.6	80.0	77.9	75.2	69.5	69.2
		MCDB	69.3	71.1	72.6	76.5	82.0	86.3	87.3	87.0	84.8	80.6	73.8	71.8
	10%	WB	64.0	66.3	67.4	70.9	74.0	78.1	78.8	79.1	77.0	73.3	67.2	67.0
		MCDB	66.6	68.8	71.1	75.2	80.7	85.1	86.2	85.8	83.6	78.5	71.0	69.8
(35) Mean Daily Temperature Range	5% DB	MDBR	20.3	20.4	21.0	20.8	20.1	16.6	15.8	16.2	17.6	21.6	23.0	19.1
		MCDBR	18.0	17.8	19.9	20.7	21.9	18.4	17.2	18.0	18.9	20.3	21.6	17.7
		MCWBR	13.9	11.5	11.4	11.3	10.2	8.0	7.4	8.0	8.4	10.8	14.2	14.1
	5% WB	MCDBR	15.7	13.7	15.2	16.4	18.0	15.5	15.9	16.0	16.6	16.7	18.4	15.9
		MCWBR	13.6	10.9	10.9	11.2	9.4	8.0	7.4	7.7	8.3	10.6	14.4	14.0
(40) Clear-Sky Solar Irradiance	taub		0.331	0.348	0.372	0.402	0.438	0.463	0.511	0.500	0.445	0.383	0.354	0.340
	taud		2.546	2.500	2.428	2.345	2.282	2.263	2.149	2.177	2.298	2.456	2.475	2.530
	Ebn at Noon		286	290	289	283	272	263	251	252	263	275	274	277
	Edh at Noon		27	31	36	40	43	43	49	47	40	32	29	26
(44) All-Sky Solar Radiation	RadAvg		969	1149	1556	1889	2092	1990	1874	1757	1650	1445	1105	851
	RadStd		75	157	123	150	109	151	131	128	100	145	96	77

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (29 neighbors)	+0.85	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+288	+187

Nomenclature: See separate page